The optimal CGLMP measurements on maximally entangled states

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For projective measurements on a prescribed maximally entangled state, we prove the optimal CGLMP value and classify every maximizer. Equality requires the local dimension to be a multiple of the number of measurement outcomes, and the measurements are Fourier-phase bases. A positive pinching identity and a tilted random partition give a deficit certificate. This resolves the first assertion of Problem 27.B in the projective class.

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Bell inequalities distinguish quantum correlations from those produced by local hidden variables [1–3]. The CGLMP family [4] uses two measurement settings per party and d outcomes. For the prescribed maximally entangled state, which projective measurements give the largest violation? Numerical searches [5, 6] found Fourier-phase measurements in every dimension tested. We prove their optimality and classify all maximizers, allowing arbitrary projection ranks and arbitrary local dimension. This resolves the first assertion of Problem 27.B for projective measurements on maximally entangled states [7, 8]. Section I states the result; Sections II–IV prove it.

I. THE CGLMP EXPRESSION AND THE THEOREM

Bell scenarios and the local bound.—Alice and Bob choose settings $x, y \in \{0, 1\}$ and obtain outcomes $a, b \in \mathbb{Z}_d$. Their behaviour is the conditional distribution $q(a, b \,|\, x, y)$. Local behaviours are convex combinations of deterministic assignments $(a_0, a_1, b_0, b_1) \in \mathbb{Z}_d^4$; these form the local polytope. For $d = 2$, recording outcomes as signs gives the CHSH inequality [2],

$$\langle A_0B_0\rangle + \langle A_1B_0\rangle + \langle A_1B_1\rangle - \langle A_0B_1\rangle \le 2. \tag{1}$$

Its d-outcome generalization uses the elementary identity

$$(a_0 - b_0) + (b_0 - a_1) + (a_1 - b_1) + (b_1 - a_0 - 1) = -1. \tag{2}$$

The CGLMP inequality.—For outcomes in $\mathbb{Z}_d$ define the residuals

$$r_{00} = [a - b]_d, \qquad r_{10} = [b - a]_d,$$
$$r_{11} = [a - b]_d, \qquad r_{01} = [b - a - 1]_d,$$

where $[u]_d \in \{0, \dots, d - 1\}$ is u mod d, and put

$$S_d(q) = \sum_{x,y\in\{0,1\}} \mathbb{E}_q\big(r_{xy}\big), \qquad I_d(q) = 4 - \frac{2S_d(q)}{d-1}. \tag{3}$$

For a deterministic assignment the sum of the four residuals is congruent, by (2), to -1 modulo d, so this sum—a sum of four representatives in $\{0, \dots, d - 1\}$—is at least $d - 1$. Hence $S_d \ge d - 1$, that is $I_d \le 2$, for every local behaviour. This is the CGLMP inequality [4], in the normalization of [9], chosen so that the local bound is 2 in every dimension; at $d = 2$, unwinding the residuals shows that I_2 is exactly the left side of (1). The CGLMP inequality is a facet of the local polytope for every d [10]. Its value alone does not determine statistical strength or resistance to an unspecified noise model; we distinguish those quantities in the Discussion.

Quantum strategies.—A *strategy* is any physical arrangement that produces a behaviour; the strategies we study are quantum. Alice and Bob share the maximally entangled state

$$\Phi_n = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} |jj\rangle$$

of two n-level systems, and each measurement is projective: it is described by a *projection-valued measure* (PVM), a family of d orthogonal projections on $\mathbb{C}^n$ summing to the identity, one per outcome, the probability of outcome a being the expectation of the ath projection. Some projections may be zero, and their ranks may differ. A *projective strategy* on Φ_n is a choice of four such PVMs, two for Alice and two for Bob. Writing A_a^x for the projection of Alice's measurement x onto outcome a, and B_b^y likewise for Bob, the behaviour is

$$q(a, b \,|\, x, y) = \langle \Phi_n | A_a^x \otimes B_b^y | \Phi_n \rangle = \tau_n\big((A_a^x)^T B_b^y\big),$$

where $\tau_n = \mathrm{Tr}/n$ is the normalized trace on $n \times n$ matrices and the transpose is taken in the basis $\{|j\rangle\}$: on the maximally

entangled state, probabilities are normalized traces of products of measurement operators. It is convenient to arrange the four PVMs in the cyclic order

$$(P^0, P^1, P^2, P^3) = (A_0^T, B_0, A_1^T, B_1), \tag{4}$$

and to write P_a^t for the projection of the PVM P^t onto outcome $a \in \mathbb{Z}_d$, so that $q(a, b \mid x, y) = \tau_n(P_a^{2x} P_b^{1+2y})$, with the four measurement contexts corresponding to cyclically adjacent pairs. Outcome labels are read modulo d throughout. We call $P^0, \ldots, P^3$ the *tracial PVMs* of the strategy. Conjugating all four by one unitary U on $\mathbb{C}^n$ corresponds to the local unitary $\overline{U} \otimes U$, which leaves Φ_n unchanged; strategies related in this way are physically equivalent.

The Fourier-phase measurements.—For $t \in \{0, 1, 2, 3\}$ and $a \in \mathbb{Z}_d$ put

$$\psi_{t,a} = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} e^{-2\pi \mathrm{i} j(a - t/4)/d} |j\rangle, \quad \Pi_a^t = |\psi_{t,a}\rangle\langle\psi_{t,a}|. \tag{5}$$

For each t the vectors $\psi_{t,0}, \ldots, \psi_{t,d-1}$ form an orthonormal basis of $\mathbb{C}^d$: the computational basis transformed by the discrete Fourier transform and then by the diagonal unitary $\mathrm{diag}(e^{\pi \mathrm{i} jt/(2d)})_j$. Hence $\Pi^t = (\Pi_a^t)_{a \in \mathbb{Z}_d}$ is a PVM for each t, and these four PVMs are the *Fourier-phase measurements*. The strategy with tracial PVMs $P_a^t = \Pi_a^t$ on Φ_d, so that Alice's two physical bases are the complex conjugates of $\psi_{0,\cdot}$ and $\psi_{2,\cdot}$ and Bob's are $\psi_{1,\cdot}$ and $\psi_{3,\cdot}$, has the CGLMP value

$$I_d(q_{\mathrm{F}}) = \frac{4}{d(d-1)} \sum_{k=1}^{d-1} (d-k) \sec \frac{\pi k}{2d} \equiv I_{\mathrm{F}}(d), \tag{6}$$

where q_{F} is the behaviour of the Fourier-phase measurements on Φ_d. The formula in (6) is Lemma 1 below with $V_j = I$. Thus $I_{\mathrm{F}}(2) = 2\sqrt{2}$, the bound of Tsirelson [11], $I_{\mathrm{F}}(3) = 2.8729\ldots$, and $I_{\mathrm{F}}(d) \to 32G/\pi^2 = 2.9698\ldots$ as $d \to \infty$, where $G = \sum_{k \geq 0} (-1)^k (2k+1)^{-2} = 0.9159\ldots$ is Catalan's constant [12, 13]. These measurements were found numerically by Kaszlikowski et al. [5] and by Durt, Kaszlikowski and Żukowski [6]; the value (6) is computed in [4]. Our result is that no other choice of projective measurements on a maximally entangled state gives a larger CGLMP value, and that any choice attaining the same value is the Fourier-phase measurements up to a local change of basis. Stated precisely we have the main theorem:

Theorem 1 (Value and rigidity)**.** *Let $d \geq 2$. Every projective strategy on Φ_d satisfies $I_d \leq I_{\mathrm{F}}(d)$. Equality holds if and only if there is a unitary U on $\mathbb{C}^d$ such that the tracial PVMs are*

$$P_a^t = U \, \Pi_a^t \, U^\dagger \qquad \textit{for all } t, a.$$

In words: the maximizers are exactly the Fourier-phase measurements, up to the local unitary $\overline{U} \otimes U$ acting on the maximally entangled state. In particular every maximizing measurement is a measurement in a rank-one basis, although nothing of the kind was assumed. Theorem 3 in Appendix D extends this to a maximally entangled state of any local dimension n but where the measurement outcomes still take one of d values. Saturation of the inequality then requires d to divide n and the maximizers are Fourier-phase measurements on a d-level subsystem tensored with the identity of an inert (n/d)-level system. Furthermore, for $d \nmid n$ the value $I_{\mathrm{F}}(d)$ is not attained by any projective strategy.

Previous work.—The value (6) was conjectured to be optimal on maximally entangled states since [6]. Lang, Vértesi and Navascués [14] proved it numerically for $d = 3$ to within 10^{-10} with a semidefinite relaxation of the tracial problem (4). Optimization over all states is a different problem: partially entangled states violate the inequality more [15], the hierarchy of [16] and the symmetry-adapted certificates of [17] give the exact all-state value for $d = 3, 4$, Yang et al. [18] certified the qutrit optimum in a black-box way, and [13, 19] treated the limit $d \to \infty$. Device-independent rigidity of the same measurements is known for Bell expressions tailored to maximally entangled states: they are self-tested in every dimension with two settings in [20, 21], and mutually unbiased bases are self-tested in [22, 23]; see the reviews [3, 24]. The Fourier-phase measurements are also the ones used in high-dimensional Bell experiments [25–27]. Mishra and Senthilkumar [28] prove the same projective value, rigidity, and divisibility statements. The comparison here uses their 20 October 2026 commit, identified in the reference. Their clock reduction uses a cyclic twirl and a finite Stone–von Neumann theorem; Lemma 1 gives explicit transition matrices and the reducing isometry. Both approaches use a positive matrix measure and a strip inequality. Here Lemma 8 gives a pinching identity and its holomorphic continuation, using the trace-density moments of Heinävaara [29, Eq. (16), proof of Lemma 2.5]. Random cells also occur in their argument; our partition law is the explicit density (29), bounded below by c_*. Their cone certificate uses interval computations for $d \leq 200$, $201 \leq d \leq 2000$, and a uniform computer-assisted estimate beyond that range. Our beta–Stein estimate treats every $d \geq 66$ analytically, leaving 2,080 scalar comparisons for $2 \leq d \leq 65$. The contribution developed here is this explicit deficit certificate and its direct forbidden-overlap route to equality classification.

Proof sketch.—Section II encodes every strategy into d unitary matrices $V_0, \ldots, V_{d-1}$ with $V_j^4 = I$, in such a way that the deficit $I_{\mathrm{F}}(d) - I_d$ becomes an explicit linear expression in their pairwise overlaps $\tau(V_j^\dagger V_k)$. The task is then to show that this expression is nonnegative for every such family of matrices, and zero for essentially one. In Section III, we use Fourier analysis to derive a family of constraints on these overlaps, and combine them to show that the deficit is bounded below by a positive multiple of a manifestly nonnegative quantity, which measures how far the eigenspaces of the different V_j are from being aligned. This proves the bound. In Section IV, we show that when the deficit vanishes, the V_j must commute and take a single form that corresponds exactly to the Fourier-phase measurements.

Conventions.—$\tau = \tau_r = \mathrm{Tr}/r$ denotes the normalized trace on $r \times r$ matrices; the size r will be clear from context. For

matrices, $\langle X, Y\rangle = \tau(X^\dagger Y)$ and $\|X\|_2^2 = \tau(X^\dagger X)$; $\|X\|$ is the operator norm. A unitary V with $V^4 = I$ has the spectral decomposition $V = \sum_{\ell\in\mathbb{Z}_4} \mathrm{i}^\ell E_\ell$ with orthogonal projections $E_0, \dots, E_3$ summing to I (some possibly zero); we call them the spectral projections of V and read their index modulo four.

II. ENCODING QUANTUM STRATEGIES INTO UNITARIES

A projective strategy is specified by its four tracial PVMs P^0, P^1, P^2, P^3 of (4), and only cyclically adjacent pairs enter the CGLMP expression: the four contexts (x, y) are the pairs (P^0, P^1), (P^1, P^2), (P^2, P^3) and (P^3, P^0). It helps to picture the four PVMs as the vertices of a square and the four contexts as its edges. Lemma 1 below trades the four PVMs for a single unitary of order four on an enlarged space—a "clock" that steps around the square—and then block-diagonalizes that unitary into d unitaries $V_0, \dots, V_{d-1}$ of order four, one for each Fourier mode j of the outcome label. All information about I_d survives in the normalized overlaps

$$z_{jk} = \tau_r(V_j^\dagger V_k), \qquad 0 \le j < k < d.$$

For a coefficient sequence $c = (c_k)_{k=1}^{d-1}$ define the quadratic form

$$L_c(V) = \sum_{j<k} \big[c_{k-j}(1 - \operatorname{Re} z_{jk}) + c_{d-k+j} \operatorname{Im} z_{jk}\big], \quad (7)$$

and let

$$\mathcal{G}_d(V) = L_\beta(V), \qquad \beta_k = \frac{1}{2d^2} \sec\frac{\pi k}{2d}, \quad (8)$$

so that explicitly

$$\mathcal{G}_d = \frac{1}{2d^2} \sum_{j<k} \Big[\sec\tfrac{\pi(k-j)}{2d}(1 - \operatorname{Re} z_{jk}) + \csc\tfrac{\pi(k-j)}{2d} \operatorname{Im} z_{jk}\Big].$$

The *forbidden overlap* of the tuple is

$$\mathfrak{F}_d(V) = \sum_{j<k} p_{jk}, \qquad p_{jk} = \frac{1 - \operatorname{Re} z_{jk} + \operatorname{Im} z_{jk}}{2}. \quad (9)$$

In terms of the spectral projections $V_j = \sum_\ell \mathrm{i}^\ell E_{j\ell}$, a short computation with $z_{jk} = \sum_{\ell,m} \mathrm{i}^{m-\ell} \tau(E_{j\ell} E_{km})$ gives

$$p_{jk} = \sum_{\ell=0}^{3} \sum_{u=1}^{2} \tau(E_{j\ell} E_{k,\ell+u}) \ge 0, \quad (10)$$

with spectral indices read modulo four: p_{jk} collects the overlaps in which the spectral label jumps by one or two units in the wrong direction. Each term is nonnegative because $\tau(EF) = \tau\big((EF)(EF)^\dagger\big) \ge 0$ for projections E, F. Thus $\mathfrak{F}_d \ge 0$, and $\mathfrak{F}_d = 0$ forces every forbidden product $E_{j\ell} E_{k,\ell+1}$ and $E_{j\ell} E_{k,\ell+2}$ to vanish, which will be used to prove uniqueness in Section IV.

The following construction retains the original PVMs on an explicit reducing subspace, which is essential for the equality classification.

Lemma 1 (Companion representation)**.** *Every projective strategy on Φ_n, $n \ge 1$, has a companion tuple $V_0, \dots, V_{d-1} \in U(4n)$ with $V_j^4 = I$ satisfying*

$$I_{\mathrm{F}}(d) - I_d = \frac{8d}{d-1} \mathcal{G}_d(V). \quad (11)$$

Moreover, the tuple determines four PVMs $\mathcal{M}^0(V), \dots, \mathcal{M}^3(V)$ on $\mathbb{C}^d \otimes \mathbb{C}^{4n}$, the companion PVMs (14)*, and an isometry $\mathcal{Z} : \mathbb{C}^n \to \mathbb{C}^d \otimes \mathbb{C}^{4n}$ such that*

$$\mathcal{M}_a^t(V)\, \mathcal{Z} = \mathcal{Z}\, P_a^t \qquad \text{for all } t, a. \quad (12)$$

In other words, on the range of $\mathcal{Z}$, the companion PVMs act exactly as the original ones.

Proof. Step 1: a random shift of the outcomes. Let $J|a\rangle = |a-1\rangle$ be the cyclic shift on $\mathbb{C}^d$, and define on $\mathbb{C}^d \otimes \mathbb{C}^n$ the operators

$$Z_t = \sum_{c=0}^{d-1} J^c \otimes P_c^t \quad (0 \le t < 4), \qquad Z_4 = Z_0(J \otimes I_n),$$

where, as above, P_c^t is the projection of P^t onto outcome c. Each Z_t is unitary, because the P_c^t are orthogonal projections summing to I_n, and commutes with $J \otimes I_n$. Let $\mathsf{C}_a^{(n)} = |a\rangle\langle a| \otimes I_n$ be the computational-basis PVM on $\mathbb{C}^d \otimes \mathbb{C}^n$. Conjugating it by Z_t gives

$$Z_t\, \mathsf{C}_a^{(n)}\, Z_t^\dagger = \sum_{k=0}^{d-1} |k\rangle\langle k| \otimes P_{a-k}^t,$$

with $a - k$ read modulo d. The four PVMs $(Z_t \mathsf{C}_a^{(n)} Z_t^\dagger)_{a\in\mathbb{Z}_d}$, $t = 0, \dots, 3$, form a projective strategy on Φ_{dn} whose outcomes are those of the original strategy shifted by a common, uniformly random $k \in \mathbb{Z}_d$: for $t \le 2$,

$$\tau_{dn}\big(Z_t \mathsf{C}_a^{(n)} Z_t^\dagger\, Z_{t+1} \mathsf{C}_b^{(n)} Z_{t+1}^\dagger\big) = \frac{1}{d} \sum_{k\in\mathbb{Z}_d} \tau_n\big(P_{a-k}^t P_{b-k}^{t+1}\big),$$

and similarly on the last edge, where the extra shift in Z_4 accounts for the -1 in the residual r_{01}. The common shift cancels from every residual r_{xy}, so I_d is unchanged, and the original strategy sits in the $k = 0$ block.

Step 2: the clock. Put $\mathsf{U}_t = Z_t^\dagger Z_{t+1}$, the transition unitary along edge t of the square, from P^t to P^{t+1}; the shift in Z_4 implements the -1 in the residual r_{01}. On a four-dimensional clock register define

$$\mathcal{R} = \sum_{t=0}^{3} |t\rangle\langle t + 1 \bmod 4| \otimes \mathsf{U}_t.$$

Every cyclic product $\mathsf{U}_t \mathsf{U}_{t+1} \mathsf{U}_{t+2} \mathsf{U}_{t+3}$ telescopes to $Z_t^\dagger Z_t (J \otimes I_n)$, so $\mathcal{R}^4 = I_4 \otimes J \otimes I_n$. Diagonalize J by the discrete Fourier transform $\mathsf{F}_{aj} = d^{-1/2} e^{2\pi \mathrm{i} aj/d}$ and pick the fourth roots $\lambda_j = e^{\pi \mathrm{i} j/(2d)}$ of its eigenvalues. Since $\mathcal{R}$ commutes

with $I_4 \otimes J \otimes I_n$, reordering the tensor factors as $\mathbb{C}^d \otimes (\mathbb{C}^4 \otimes \mathbb{C}^n)$ and Fourier transforming the outcome coordinate gives

$$\mathcal{R} = (\mathsf{F} \otimes I_{4n})\ \mathrm{diag}(\lambda_j V_j)_{j=0}^{d-1}\ (\mathsf{F}^\dagger \otimes I_{4n}) \tag{13}$$

with $V_j \in U(4n)$; taking fourth powers shows $V_j^4 = I$. In these coordinates define the companion PVMs on $\mathbb{C}^d \otimes \mathbb{C}^{4n}$,

$$\mathcal{M}_a^t(V) = \mathcal{R}^t\, \mathsf{C}_a^{(4n)}\, \mathcal{R}^{-t}, \qquad 0 \le t < 4. \tag{14}$$

Their value of I_d on Φ_{4dn} equals the original: the edge distribution $\tau(\mathcal{M}_a^t \mathcal{M}_b^{t+1}) = \tau(\mathsf{C}_a^{(4n)} \mathcal{R}\, \mathsf{C}_b^{(4n)} \mathcal{R}^\dagger)$ is independent of t, and tracing out the clock averages the four transition unitaries U_t, so each companion edge carries one quarter of the four original edge costs.

The original strategy survives inside: in clock position zero, $\mathsf{U}_0 \cdots \mathsf{U}_{t-1} = Z_0^\dagger Z_t$, so the isometry

$$\mathcal{Z}\xi = \sum_a |a\rangle \otimes |0\rangle \otimes P_a^0 \xi \tag{15}$$

satisfies $\mathcal{Z}^\dagger \mathcal{Z} = I_n$ and $\mathcal{M}_a^t(V)\mathcal{Z} = \mathcal{Z} P_a^t$, which is (12). Since the $\mathcal{M}_a^t$ are self-adjoint, both the range of $\mathcal{Z}$ and its orthogonal complement are invariant under every $\mathcal{M}_a^t$, and on the range the companion PVMs are a copy of the original ones.

Step 3: the cost in terms of overlaps. Write $\omega = e^{2\pi \mathrm{i}/d}$. By (13), the block of $\mathcal{R}$ between computational levels a and b of the outcome coordinate is $R_{ab} = d^{-1} \sum_j \omega^{j(a-b)} \lambda_j V_j$, which depends only on $a - b$. The probability that the outcomes of one edge differ by c is therefore

$$\Pr(c) = \frac{1}{d^2} \sum_{j,k=0}^{d-1} \omega^{c(j-k)} \lambda_j \bar{\lambda}_k\, \tau_{4n}(V_k^\dagger V_j).$$

Summing $\sum_c c\, \omega^{c(j-k)} = -d/(1-\omega^{j-k})$ for $j \not\equiv k \pmod d$ and $\sum_c c = d(d-1)/2$ for $j = k$, and using, for $x = (j-k)/d \in (0,1)$,

$$\begin{aligned} \frac{\lambda_j \bar{\lambda}_k}{1-\omega^{j-k}} &= \frac{e^{\pi \mathrm{i} x/2}}{2}\,(1 + \mathrm{i}\cot \pi x) \\ &= \frac{1}{4}\Big(\sec\frac{\pi x}{2} + \mathrm{i}\csc\frac{\pi x}{2}\Big), \end{aligned}$$

the expected residual of one edge becomes

$$\begin{aligned} \frac{d-1}{2} - \frac{1}{2d} \sum_{j<k} \Big[&\sec\tfrac{\pi(k-j)}{2d}\, \mathrm{Re}\, z_{jk} \\ &- \csc\tfrac{\pi(k-j)}{2d}\, \mathrm{Im}\, z_{jk} \Big]. \end{aligned} \tag{16}$$

Subtracting the value of (16) at $V_j = I$ gives exactly $d\,\mathcal{G}_d(V)$. Summing the four edges, S_d exceeds its value at $V_j = I$ by $4d\,\mathcal{G}_d$, and (3) turns this into (11), provided the strategy at $V_j = I$ has value $I_{\mathrm{F}}(d)$. Indeed, put $R_0 = \mathsf{F}\,\mathrm{diag}(\lambda_j)\mathsf{F}^\dagger$ and $C_a = |a\rangle\langle a|$. At $V_j = I$, the companion unitary is $\mathcal{R}_0 = R_0 \otimes I_{4n}$. Since $\mathsf{F}^\dagger|a\rangle = \psi_{0,a}$ and $\mathrm{diag}(\lambda_j^t)\psi_{0,a} = \psi_{t,a}$, one has

$$R_0^t C_a R_0^{-t} = \mathsf{F} \Pi_a^t \mathsf{F}^\dagger. \tag{17}$$

Thus the companion PVMs are a common Fourier conjugate of the standard amplification. Their probabilities agree, and evaluating (16) at $z_{jk} = 1$ and inserting into (3) yields (6). □

III. BOUNDING $\mathcal{G}_d(V)$

Section II reduced Theorem 1 to a statement about d unitaries $V_0, \dots, V_{d-1}$ of order four: the quantity $\mathcal{G}_d(V)$, a combination of their overlaps z_{jk} with secant and cosecant coefficients, must be nonnegative. We obtain inequalities between the overlaps from the following construction. Cut the unit circle, parametrized by $t \in [0,1)$, into d consecutive arcs, put $V(t) = V_j$ on the jth arc, and form the matrix-valued function

$$U(e^{2\pi \mathrm{i} t}) = e^{-\pi \mathrm{i} t/2}\, V(t)^\dagger.$$

Because $V_j^4 = I$, this function satisfies $U(z)^4 = z^{-1} I$ at every point $z = e^{2\pi \mathrm{i} t}$ of the circle. (The phase $e^{-\pi \mathrm{i} t/2}$ is the continuous version of the fourth roots $\lambda_j = e^{\pi \mathrm{i} j/(2d)}$ in (13): at $t = j/d$, U equals $(\lambda_j V_j)^\dagger$.) We study U through its Fourier series.

Write $U(z) = \sum_{k \in \mathbb{Z}} c_k z^k$ in L^2, with

$$c_k = \int_0^1 U(e^{2\pi \mathrm{i} t}) e^{-2\pi \mathrm{i} k t}\, \mathrm{d}t.$$

For the negative-frequency projection $P_- U = \sum_{k<0} c_k z^k$, Parseval's identity gives

$$\|P_- U\|_2^2 = \sum_{k<0} \tau_r(c_k^\dagger c_k).$$

A unitary-valued U has $\|U\|_2^2 = 1$.

We are interested in unitary-valued U with $U(z)^4 = z^{-1} I$, which we call *fourth roots of* z^{-1}. Discontinuous ones are allowed and necessary, since the functions built above from the V_j have jumps at the partition boundaries. The basic example has $r = 1$: $U_0(e^{2\pi \mathrm{i} t}) = e^{-\pi \mathrm{i} t/2}$ for $0 \le t < 1$, which jumps by a factor $-\mathrm{i}$ when t passes 1. Its coefficients satisfy $|c_k|^2 = 8/[\pi^2(4k+1)^2]$. Using $\sum_{n \ge 0}(4n+3)^{-2} = \pi^2/16 - G/2$, where $G = \sum_{k \ge 0}(-1)^k (2k+1)^{-2} = 0.9159\ldots$ is Catalan's constant from (6), we find

$$\|P_- U_0\|_2^2 = \frac{8}{\pi^2} \sum_{n \ge 0} \frac{1}{(4n+3)^2} = \frac{1}{2} - \frac{4G}{\pi^2} =: \nu_4. \tag{18}$$

The next lemma says that no fourth root of z^{-1}, of any matrix size, has a smaller value of $\|P_- U\|_2^2$.

Lemma 2 (Negative frequencies of fourth roots)**.** *Let U be a measurable function from the unit circle to the $r \times r$ unitary matrices with $U(z)^4 = z^{-1} I$ for almost every z. Then*

$$\|P_- U\|_2^2 \ge \nu_4. \tag{19}$$

The proof is given in Appendix A, together with an explanation of the idea. In brief, $1 - 2\|P_- U\|_2^2$ is rewritten as the average over the circle of $\tau_r(A\, \mathsf{H} B)$, where A and B are commuting Hermitian matrices with eigenvalues ± 1 built from U, and $\mathsf{H}B$ is obtained from B by multiplying its kth Fourier coefficient by $-\mathrm{i}\,\mathrm{sgn}\, k$. The average is then bounded using an

analytic function h on the strip $|\mathrm{Re}\, z| < 1$ whose real part equals $|y|$ on the two boundary lines; the constant $8G/\pi^2 = 1 - 2\nu_4$ is the value of h at the centre of the strip, and the average of an analytic function over the circle is its value at the centre. When B and $\mathsf{H}B$ commute this is the whole argument. In general they do not, and the new ingredient is a comparison identity for noncommuting matrices.

We now apply Lemma 2 to the function

$$U(e^{2\pi \mathrm{i}t}) = e^{-\pi \mathrm{i}t/2} V(t)^\dagger$$

built from V and a partition of the circle. The result is a quadratic form of the type L_c in (7), and to write down its coefficients we need two pieces of notation. First, let

$$\sigma(x) = \sec \frac{\pi x}{2} \qquad (0 \le x < 1),$$

so that $\sigma(1-x) = \csc(\pi x/2)$ and the coefficients of $\mathcal{G}_d$ in (8) are $\beta_k = \sigma(k/d)/(2d^2)$, $1 \le k < d$. Second, for a function g on $[0,1]$ and an integer $0 \le k \le d$, let

$$P_{d,k}(x) = \binom{d}{k} x^k (1-x)^{d-k},$$
$$(B_d g)(k) = d \int_0^1 P_{d,k}(x)\, g(x)\, \mathrm{d}x.$$

The weight $P_{d,k}(x)$ is the probability that a binomial variable with parameters (d, x) equals k; as a function of x it is concentrated near $x = k/d$ and has integral $1/(d+1)$. Thus $(B_d g)(k)$ is a weighted average of g over a neighbourhood of the point k/d, normalized up to the factor $d/(d+1)$. We call these Bernstein averages, since the weights are Bernstein basis polynomials [30]; B_d is not the classical Bernstein approximation operator. Finally, let

$$q_k = \frac{(B_d \sigma)(k)}{2d^2}, \qquad 1 \le k < d, \tag{20}$$

which is the coefficient β_k of $\mathcal{G}_d$ with the point value $\sigma(k/d)$ replaced by its Bernstein average, and let L_q be the form (7) with the coefficients $c_k = q_k$.

Lemma 3 (Partition forms)**.** *Let $\ell = (I_0, \dots, I_{d-1})$ be a partition of $[0,1]$ into consecutive intervals, and let $V(t) = V_j$ for $t \in I_j$. Then*

$$\begin{aligned}\mathcal{K}_\ell(V) &:= \|P_-[e^{-\pi \mathrm{i}t/2} V(t)^\dagger]\|_2^2 - \nu_4 \\ &= \frac{1}{2} \sum_{j<k} \int_{I_j}\int_{I_k} \big[\sigma(t-s)(1 - \mathrm{Re}\, z_{jk}) \\ &\qquad + \sigma(1-t+s)\, \mathrm{Im}\, z_{jk}\big]\, \mathrm{d}t\, \mathrm{d}s \;\ge\; 0.\end{aligned} \tag{21}$$

Extend $V_{j+d} = -\mathrm{i}V_j$. For a circular partition with d boundary points, averaging $\mathcal{K}$ over the d cyclic assignments of $V_0, \dots, V_{d-1}$ to the arcs gives L_c in (7) *with*

$$c_k = \frac{1}{2d} \int_0^1 \sigma(x)\, \Pr(N_x = k)\, \mathrm{d}x, \qquad 1 \le k < d, \tag{22}$$

where N_x counts boundary points in an arc of length x from a uniformly random origin. In particular, taking the d boundary points independent and uniform gives $\Pr(N_x = k) = P_{d,k}(x)$ and hence $L_q(V) \ge 0$.

Proof. The function $U(e^{2\pi \mathrm{i}t}) = e^{-\pi \mathrm{i}t/2} V(t)^\dagger$ satisfies $U(z)^4 = z^{-1} I$, so Lemma 2 gives $\mathcal{K}_\ell \ge 0$. For the formula, expand $\|P_- U\|_2^2 = \langle U, P_- U\rangle$ with the Abel-regularized kernel of P_-, whose limit at argument $x = t - s$ is $-\frac{1}{2}(1 + \mathrm{i}\cot \pi x)$, and subtract the same expansion at $V_j = I$, which equals ν_4 by (18). The diagonal blocks $I_j \times I_j$ cancel in the difference. A cross block contributes the kernel times $\tau(V(t)^\dagger V(s)) e^{\pi \mathrm{i}(t-s)/2} - e^{\pi \mathrm{i}(t-s)/2}$; combining the block $I_j \times I_k$ with its transpose and using, for $0 < x < 1$,

$$e^{\pi \mathrm{i}x/2}(1 + \mathrm{i}\cot \pi x) = \frac{\sigma(x) + \mathrm{i}\,\sigma(1-x)}{2},$$

the two half-squares assemble into the real integrand of (21). The integrals are finite. Near $t - s = 1$, the cross-sectional measure is at most $1 - (t-s)$, which cancels the pole of $\sigma(t-s)$. For adjacent intervals, the region $0 < t - s \le x$ is contained in a triangle of area $x^2/2$; for separated intervals it is empty for small x. Thus its area is $O(x^2)$, and the kernel $\sigma(1-t+s) = O((t-s)^{-1})$ is integrable. These bounds also justify the Abel-regularization limit.

For the circular average, first note that in one dimension a single drop of the phase label by one power of i costs nothing: for $a \in \mathbb{Z}_4$ and a drop at T,

$$e^{-\pi \mathrm{i}t/2}\, \mathrm{i}^{-a + \mathbf{1}_{\{t>T\}}} = \mathrm{i}^{1-a} e^{-\pi \mathrm{i}T/2}\, U_0([t-T]_1) \tag{23}$$

almost everywhere; translation and a constant phase preserve the moduli of the Fourier coefficients, so the negative-frequency part of this function has squared norm exactly ν_4. Consequently $\mathcal{K} = 0$ whenever all V_j are equal multiples of the identity in the extended sequence, and the circular form, with the value at $V_j = I$ subtracted, is again a combination of the quantities $1 - \mathrm{Re}\, z_{jk}$ and $\mathrm{Im}\, z_{jk}$. For a pair of arcs with labels $j < k$ separated by $k - j$ boundary points along the direct route, the kernel computation above assigns the coefficient $\sigma(x)$ to the direct arc and $\sigma(1-x)$ to the complementary arc. If the pair straddles the parameter origin, the extension $V_{j+d} = -\mathrm{i}V_j$ replaces z_{jk} by $-\mathrm{i}\bar{z}_{jk}$, which interchanges the two coefficients—precisely as the direct route then runs through the complementary arc. Averaging the starting label over the d cyclic assignments and integrating over a uniform origin yields L_c with (22). Explicitly, put $m = k - j$ and $P_m(x) = \Pr(N_x = m)$. Before simplification, the coefficient of $1 - \mathrm{Re}\, z_{jk}$ is

$$\frac{1}{2d} \int_0^1 (1-x)\big[\sigma(x) P_m(x) + \sigma(1-x) P_{d-m}(x)\big]\, \mathrm{d}x.$$

Complementary arcs obey $P_{d-m}(x) = P_m(1-x)$. Substituting $x \mapsto 1 - x$ in the second term combines the weights $1 - x$ and x, giving $c_m = (2d)^{-1} \int_0^1 \sigma(x) P_m(x)\, \mathrm{d}x$. For independent uniform boundary points, N_x is binomial with parameters (d, x), so $\Pr(N_x = k) = P_{d,k}(x)$ and $c_k = q_k$; nonnegativity of each $\mathcal{K}$ gives $L_q \ge 0$. □

The uniform average L_q does not match $\mathcal{G}_d$: Bernstein averaging flattens the secant. The next lemma corrects this by giving the boundary points a non-uniform probability law. Define

$$v(x) = -\frac{x(1-x)}{2} + \frac{2x-1}{\pi\big[\sec\frac{\pi x}{2} - \csc\frac{\pi x}{2}\big]}, \tag{24}$$

with removable singularities filled continuously, and

$$b(x) = -\frac{1}{2}\Big[x(1-x)\sigma(x) - \frac{2x}{\pi}\Big]'',$$
$$S(x) = \frac{1 - z\cot z}{\sin^2 z}, \qquad z = \frac{\pi}{2}\Big(x - \frac{1}{2}\Big), \qquad S(1/2) = \frac{1}{3}.$$

The factor $1-x$ cancels the pole of $\sigma(x)$ at $x = 1$. Subtracting $2x/\pi$ removes the regularized endpoint value, leaving a smooth function that vanishes at both endpoints; in particular b is bounded on $[0, 1]$. Finally put

$$e_{d,k} = \sigma(k/d) - (B_d\sigma)(k) - \frac{1}{d}(B_d b)(k),$$
$$s_{d,k} = \frac{1}{d}(B_d S)(k). \tag{25}$$

Lemma 4 (A non-uniform random partition)**.** *There is a probability density on circular partitions whose average $\mathcal{K}_d^v$ of the partition forms $\mathcal{K}$ is the form* (7) *with coefficients*

$$c_k = \frac{1}{2d^2}\Big[(B_d\sigma)(k) + \frac{1}{d}\big(B_d(v\sigma)''\big)(k)\Big], \tag{26}$$

and satisfies $\mathcal{K}_d^v \geq c_* L_q$ with $c_* = \frac{3}{2} - \frac{\pi}{4}$. Moreover,

$$\mathcal{G}_d = \mathcal{K}_d^v + \frac{1}{2d^2}L_{e+s}, \qquad s_{d,k} = s_{d,d-k} \geq \frac{1}{3(d+1)}, \tag{27}$$

where L_{e+s} is the form (7) *with coefficients $c_k = e_{d,k} + s_{d,k}$.*

Proof. The function v is symmetric about $1/2$, has zero value and first derivative at both endpoints, and $v''(0) = v''(1) = (\pi - 2)/2$. It has the expansion

$$v(x) = \sum_{n\geq 1} a_n \frac{\sin^2(\pi n x)}{\pi^2 n^2}, \quad a_n > 0, \quad \sum_{n\geq 1} a_n = \frac{\pi}{4} - \frac{1}{2}. \tag{28}$$

To prove this, write $u = x - \frac{1}{2}$, so that

$$v(x) = \frac{u^2 - \frac{1}{4}}{2} + \frac{u}{\sqrt{2}\,\pi}\big[\csc(\pi u/2) - 2\sin(\pi u/2)\big].$$

The nonconstant even Taylor coefficients of $u\csc(\pi u/2)$ are positive, and for $a = \pi/2$ and $|au| \leq \pi/4$,

$$-[u\sin(au)]^{(4)} = a^3[4\cos(au) - au\sin(au)] > 0,$$

so $v^{(4)} > 0$ on $(0, 1)$. Four integrations by parts then give

$$a_n = \frac{1}{4\pi^2 n^2}\int_0^1 v^{(4)}(x)[1 - \cos(2\pi n x)]\,\mathrm{d}x > 0,$$

with $a_n = O(n^{-2})$; Fourier inversion and the value $v''(0)$ fix the sum in (28).

Replace the uniform law of the independent boundary points $T_1, \ldots, T_d$ by the law with density, relative to the uniform one,

$$\varpi_d = 1 + \frac{1}{d}\sum_{a<b} v''(T_a - T_b)$$
$$= c_* + \frac{1}{d}\sum_{n\geq 1} a_n\Big|\sum_{j=1}^{d} e^{2\pi i n T_j}\Big|^2, \tag{29}$$

using the periodic extension of v in the differences; the second form follows from $v''(x) = 2\sum_n a_n\cos(2\pi n x)$ and (28). Its expectation is one, and $\varpi_d \geq c_* > 0$ pointwise. For a single Fourier mode $v_n = \sin^2(\pi n x)/(\pi^2 n^2)$, conditioning on two boundary points shows that the generating function of the arc count under this law is

$$(1 - x + xz)^d + (d-1)\,v_n(x)\,(z-1)^2(1 - x + xz)^{d-2},$$

and $(z-1)^2(1-x+xz)^{d-2}$ generates exactly $P''_{d,k}(x)/[d(d-1)]$. Summing the modes with (28) gives

$$\Pr_{\varpi_d}(N_x = k) = P_{d,k}(x) + \frac{v(x)}{d}P''_{d,k}(x). \tag{30}$$

Insert (30) into (22) and integrate by parts twice: the endpoint terms vanish because $v\sigma$ vanishes at both endpoints, $(v\sigma)'$ has finite one-sided limits, and $P_{d,k}$ vanishes there. This gives (26). The inequality $\mathcal{K}_d^v \geq c_* L_q$ holds because the average under ϖ_d of the pointwise-nonnegative forms $\mathcal{K}$ is bounded below by c_* times the uniform average, which is L_q.

Finally, differentiating (24) gives $(v\sigma)'' = b - S$, so the coefficient of the difference $\mathcal{G}_d - \mathcal{K}_d^v$ at separation k is $[e_{d,k} + s_{d,k}]/(2d^2)$, which is (27). The symmetry $S(x) = S(1-x)$ and $P_{d,k}(x) = P_{d,d-k}(1-x)$ give $s_{d,k} = s_{d,d-k}$, and the bounds $\sin^2 z \leq z^2$ and $1 - z\cot z \geq z^2/3$ (a positive Taylor series) give $S \geq 1/3$, hence $s_{d,k} \geq \frac{1}{3}\cdot\frac{1}{d}(B_d 1)(k) = \frac{1}{3(d+1)}$, using $\int_0^1 P_{d,k} = 1/(d+1)$. □

By (27), the deficit form $\mathcal{G}_d$ dominates $c_* L_q$ up to the error term $L_{e+s}/(2d^2)$. The error coefficients $e_{d,k}$ are small but not symmetric in $k \leftrightarrow d-k$, so L_e is not simply a combination of the nonnegative quantities p_{jk}: by (7),

$$L_e = 2\sum_{j<k} e_{d,k-j}\,p_{jk} + \sum_{j<k} t_{k-j}\,\mathrm{Im}\,z_{jk},$$

where $t_k = e_{d,d-k} - e_{d,k}$, and the skew part multiplies bare imaginary parts. The next lemma controls those through triangle "circulations", which are themselves dominated by the forbidden overlaps.

Lemma 5 (Circulation bound)**.** *For $i < j < k$, let $C_{ijk} = \mathrm{Im}\,z_{ij} + \mathrm{Im}\,z_{jk} - \mathrm{Im}\,z_{ik}$. Then*

$$|C_{ijk}| \leq 5\,(p_{ij} + p_{jk} + p_{ik}). \tag{31}$$

Moreover, with $\omega_{ijk} = t_{j-i} + t_{k-j} - t_{k-i}$,

$$L_e = 2\sum_{j<k} e_{d,k-j}\,p_{jk} + \frac{1}{d}\sum_{i<j<k} \omega_{ijk}\,C_{ijk}. \tag{32}$$

Proof. The sum-of-squares identity (B2) in Appendix B proves (31). For (32), compare the coefficients of $\operatorname{Im} z_{ab}$, $k = b-a$, on both sides. The triangles containing the edge (a, b) are (a, b, c) with $c > b$, (c, a, b) with $c < a$, and (a, c, b) with $a < c < b$; they contribute $\omega_{abc} + \omega_{cab} - \omega_{acb}$ to the coefficient. Each contains the term $+t_k$, in total $(d-2)t_k$. The remaining terms are partial sums $\mathsf{S}(N) = \sum_{m=1}^{N} t_m$: they add up to

$$2t_k + [\mathsf{S}(d-1-b) - \mathsf{S}(d-1-a)] + [\mathsf{S}(a) - \mathsf{S}(b)],$$

and the skew symmetry $t_{d-m} = -t_m$ gives $\mathsf{S}(d-1-a) - \mathsf{S}(d-1-b) = \sum_{m=d-b}^{d-1-a} t_m = -\sum_{m=a+1}^{b} t_m = \mathsf{S}(a) - \mathsf{S}(b)$, so the two brackets cancel. The total coefficient is $d\,t_k$, matching the left side. □

Two consequences. First, a coarse bound: each edge lies in $d-2$ triangles, so with $b_d = \max_k |e_{d,k}|$ and $|\omega_{ijk}| \le 6b_d$,

$$L_e \ge -2b_d\mathfrak{F}_d - \frac{6b_d}{d} \cdot 5\,(d-2)\mathfrak{F}_d \ge -32\,b_d\mathfrak{F}_d. \tag{33}$$

Second, a sharper per-edge bound: define the absolute triangle load at separation k,

$$\ell_{d,k} = \sum_{j=1}^{k-1} |t_j + t_{k-j} - t_k| + \sum_{j=k+1}^{d-1} |t_k + t_{j-k} - t_j|, \tag{34}$$

which collects $|\omega|$ over the triangles containing one edge of separation k. Then (31) and (32) give

$$L_e \ge 2 \sum_{j<k} \Big[e_{d,k-j} - \frac{5}{2d}\,\ell_{d,k-j} \Big] p_{jk}. \tag{35}$$

It remains to know that the error coefficients $e_{d,k}$—explicit numbers depending only on d and k—are small enough. That is the content of the following lemma, whose proof—a Stein-type estimate for the Bernstein averages at large d, and an exact interval computation at small d—is given in Appendix C.

Lemma 6 (Bounds on the error coefficients)**.** *(i) For every $d \ge 66$,*

$$b_d = \max_{1\le k<d} |e_{d,k}| \le \frac{45}{32(d+1)(d+2)}. \tag{36}$$

(ii) For every $2 \le d \le 65$ and every $1 \le k < d$,

$$\frac{1}{3(d+1)} + e_{d,k} - \frac{5}{2d}\,\ell_{d,k} \ge \frac{1}{6(d+1)}. \tag{37}$$

Theorem 2 (Deficit certificate)**.** *Let $d \ge 2$, $r \ge 1$, and let $V_0, \dots, V_{d-1} \in U(r)$ satisfy $V_j^4 = I$. Then $L_q(V) \ge 0$ and*

$$\mathcal{G}_d(V) \ge c_* L_q(V) + \eta_d \mathfrak{F}_d(V), \tag{38}$$

where

$$\eta_d = \begin{cases} [6d^2(d+1)]^{-1}, & 2 \le d \le 65, \\ (2d-131)\,[6d^2(d+1)(d+2)]^{-1}, & d \ge 66, \end{cases}$$

is positive in every dimension.

Proof. Lemma 3 gives $L_q \ge 0$. By Lemma 4, $\mathcal{G}_d \ge c_* L_q + L_{e+s}/(2d^2)$, so it suffices to bound $L_{e+s} \ge 2d^2\eta_d\mathfrak{F}_d$. The symmetric part is immediate: since $s_{d,k} = s_{d,d-k}$, definition (7) gives $L_s = 2\sum_{j<k} s_{d,k-j}\,p_{jk} \ge \frac{2}{3(d+1)}\mathfrak{F}_d$ by (27). For $2 \le d \le 65$, combine this with (35) and Lemma 6(ii):

$$L_{e+s} \ge 2 \min_k \Big[\frac{1}{3(d+1)} + e_{d,k} - \frac{5}{2d}\ell_{d,k} \Big] \mathfrak{F}_d \ge \frac{\mathfrak{F}_d}{3(d+1)}.$$

For $d \ge 66$, use instead (33) and Lemma 6(i):

$$\begin{aligned} L_{e+s} &\ge \Big[\frac{2}{3(d+1)} - \frac{45}{(d+1)(d+2)} \Big] \mathfrak{F}_d \\ &= \frac{2d-131}{3(d+1)(d+2)}\,\mathfrak{F}_d > 0. \end{aligned}$$

Dividing by $2d^2$ gives (38). □

Combining Lemma 1 with Theorem 2 and the fact that $\mathfrak{F}_d \ge 0$ proves the inequality of Theorem 1, and in fact for every local dimension: every projective strategy on Φ_n, in particular on Φ_d, has $I_{\mathrm{F}}(d) - I_d = \frac{8d}{d-1}\mathcal{G}_d(V) \ge 0$.

IV. PROOF THAT THE FOURIER-PHASE MEASUREMENTS UNIQUELY SATURATE THE BOUND

Proof of Theorem 1. The fact that any strategy obeys $I_d \le I_{\mathrm{F}}(d)$ was proved in the previous section and is the first part of Theorem 1. What remains is to show that if $I_d = I_{\mathrm{F}}(d)$ for a given strategy, then the projectors of that strategy are the Fourier-phase measurements given in (5) up to local unitary rotations.

To that end, suppose that $I_d = I_{\mathrm{F}}(d)$. We keep the local dimension n general through Steps 1–3, which make no use of $n = d$ and are reused in Appendix D; only Step 4 specializes. Under equality, $\mathcal{G}_d(V) = 0$ by (11), so $\mathfrak{F}_d(V) = 0$ by Theorem 2, and by (10) every forbidden product vanishes:

$$E_{j\ell}E_{k,\ell+u} = 0 \qquad (j < k,\ \ell \in \mathbb{Z}_4,\ u \in \{1,2\}), \tag{39}$$

since a projection product with $\tau(EF) = \tau[(EF)(EF)^\dagger] = 0$ is itself zero.

Step 1: the tuple commutes. Fix $j < k$. By (39), $E_{j\ell}E_{km} \ne 0$ requires $m \in \{\ell, \ell-1\}$, and taking adjoints, $E_{km}E_{j\ell'} \ne 0$ requires $m \in \{\ell', \ell'-1\}$. For $\ell \ne \ell'$ the two sets $\{\ell, \ell-1\}$ and $\{\ell', \ell'-1\}$ have at most one element in common, so at most one value of m can make $E_{j\ell}E_{km}E_{j\ell'}$ nonzero. Inserting $\sum_m E_{km} = I$,

$$0 = E_{j\ell}E_{j\ell'} = \sum_{m\in\mathbb{Z}_4} E_{j\ell}E_{km}E_{j\ell'} \qquad (\ell \ne \ell'),$$

and a sum with at most one nonzero term that equals zero has none. Hence $E_{j\ell}E_{km}E_{j\ell'} = 0$ for all m whenever $\ell \ne \ell'$. Inserting $\sum_\ell E_{j\ell} = I$ on both sides of E_{km} therefore gives

$$E_{km} = \sum_{\ell,\ell'} E_{j\ell}E_{km}E_{j\ell'} = \sum_{\ell} E_{j\ell}E_{km}E_{j\ell},$$

which commutes with every $E_{j\ell}$. Applying this to every pair $j < k$ makes all the spectral projections $E_{j\ell}$, and hence all V_j, simultaneously diagonalizable.

Step 2: the tuple is a direct sum of cuts. Decompose $\mathbb{C}^{4n}$ into joint eigenspaces, on each of which every V_j acts as a single phase i^{ℓ_j}. By (39), for $j < k$ the labels satisfy $\ell_k \in \{\ell_j, \ell_j - 1\}$: along $j = 0, \dots, d-1$ the label is constant except for at most one decrease by one. On each joint eigenspace, therefore,

$$V_j = \mathrm{i}^{\,a - \mathbf{1}_{\{j > j_0\}}}, \qquad a \in \mathbb{Z}_4, \qquad 0 \le j_0 < d, \tag{40}$$

a configuration we call a *cut*.

Step 3: every cut gives the standard measurements. For $s = d - 1 - j_0$, the phases $\lambda_j = e^{\pi \mathrm{i} j/(2d)}$ satisfy

$$\lambda_j \, \mathrm{i}^{-\mathbf{1}_{\{j > j_0\}}} = \lambda_s^{-1} \lambda_{[j+s]_d},$$

as one checks separately for $j \le j_0$ and $j > j_0$. Thus the diagonal $\mathrm{diag}(\lambda_j V_j)$ of a cut equals the standard diagonal $\mathrm{diag}(\lambda_j)$ with its frequencies shifted by s, times the global phase $\mathrm{i}^a \lambda_s^{-1}$. With $D_s = \mathrm{diag}(\omega^{bs})_{b \in \mathbb{Z}_d}$, the shift has the orientation

$$\mathsf{F}\,\mathrm{diag}(\lambda_{[j+s]_d})\mathsf{F}^\dagger = D_s^\dagger R_0 D_s. \tag{41}$$

Each D_s commutes with every C_b, and the global phase $\mathrm{i}^a \lambda_s^{-1}$ cancels from (14). Consequently, on a joint eigenspace belonging to this cut, the common intertwiner with the standard PVMs is $W_s = D_s^\dagger \mathsf{F}$, by (17). Taking these intertwiners on the joint eigenspaces gives a unitary W on $\mathbb{C}^d \otimes \mathbb{C}^{4n}$ such that

$$\mathcal{M}_b^t(V) = W(\Pi_b^t \otimes I_{4n})W^\dagger \qquad \text{for all } t, b.$$

Step 4: descent to the original space. To see that the projectors Π_a^t generate $M_d(\mathbb{C})$, first conjugate all four PVMs by F. By (17), their first basis is computational and their second consists of the columns of $U_{\mathrm{F}} = R_0 = \mathsf{F}\,\mathrm{diag}(\lambda_j)\mathsf{F}^\dagger$. An operator commuting with the first PVM is diagonal, and U_{F} has entries

$$(U_{\mathrm{F}})_{ab} = \frac{1 - \mathrm{i}}{d\,[1 - e^{2\pi \mathrm{i}(a-b+1/4)/d}]},$$

all nonzero, so a diagonal operator commuting with the rank-one projectors onto its columns has all diagonal entries equal. The commutant of the amplification $\{\Pi_a^t \otimes I_{4n}\}$ is therefore $I_d \otimes M_{4n}(\mathbb{C})$: every projection commuting with all four PVMs is $I_d \otimes p$, its rank is divisible by d, and the restriction of the PVMs to its range is again a standard amplification.

Apply this to the range projection $\Lambda = \mathcal{Z}\mathcal{Z}^\dagger$ of the isometry (15), which commutes with every $\mathcal{M}_a^t(V)$ by (12): thus $W^\dagger \Lambda W = I_d \otimes p$ for a projection p on $\mathbb{C}^{4n}$. Now set $n = d$. Comparing ranks, $d = \mathrm{rank}\,\Lambda = d\,\mathrm{rank}\,p$ forces $p = vv^\dagger$ for a unit vector $v \in \mathbb{C}^{4d}$, and $T = (I_d \otimes v^\dagger)W^\dagger \mathcal{Z}$ is a unitary on $\mathbb{C}^d$ with

$$T P_a^t T^\dagger = (I_d \otimes v^\dagger)(\Pi_a^t \otimes vv^\dagger)(I_d \otimes v) = \Pi_a^t,$$

which is the classification claimed in Theorem 1 with $U = T^\dagger$.

Conversely, if $P_a^t = U \Pi_a^t U^\dagger$, the behaviour coincides with that of the standard strategy, since $\tau_d(U\Pi U^\dagger\, U\Pi' U^\dagger) = \tau_d(\Pi\Pi')$, and the standard strategy attains $I_{\mathrm{F}}(d)$ by Lemma 1 at $V_j = I$. □

V. DISCUSSION

For projective measurements on a prescribed maximally entangled state, maximal CGLMP violation determines the Fourier-phase measurements and their multiplicity. This settles the first assertion of Problem 27.B in the projective class. Extending the argument to arbitrary d-outcome POVMs remains a separate problem in general d. Optimization over states is also different: partially entangled states can give a larger value [15].

The certificate proves the quantitative estimate

$$\mathfrak{F}_d(V) \le \frac{d-1}{8d\,\eta_d}\big(I_{\mathrm{F}}(d) - I_d\big). \tag{42}$$

This controls forbidden overlaps in the companion representation. A bound on the distance between measurement families would additionally require an approximate commuting-cut decomposition, descent through the reducing isometry, and control of constants as n varies. Neither such a measurement-distance bound nor a lower bound matching the $O(s/n)$ construction in Appendix D is proved here.

Noise resistance must specify both the noise and the witness. For uniform behaviour noise $q_v = vq + (1-v)q_{\mathrm{unif}}$, with $q_{\mathrm{unif}}(a, b \,|\, x, y) = d^{-2}$, one has

$$I_d(q_{\mathrm{unif}}) = 0, \qquad I_d(q_v) = v I_d(q).$$

For $I_d(q) > 2$, the threshold for violation of this fixed CGLMP inequality is therefore $v_{\mathrm{CGLMP}} = 2/I_d(q)$. The maximizing strategies minimize this threshold. For rank-one PVMs on Φ_d, white noise on the state induces exactly q_{unif}; for unequal projection ranks the induced noise need not be uniform. By contrast, the threshold against the entire local polytope $\mathcal{L}_d$ is

$$v_{\mathcal{L}}(q) = \max\{v \in [0,1] : vq + (1-v)q_{\mathrm{unif}} \in \mathcal{L}_d\}.$$

Mishra and Senthilkumar [28, `clauses/noise-literal`, Theorem 1] give better values of $v_{\mathcal{L}}$ for $d \ge 4$ using projective measurements with unused outcomes. This does not contradict optimality for the fixed CGLMP witness or establish a counterexample for rank-one PVMs on Φ_d. Kullback–Leibler statistical strength, defined by relative entropy to local behaviours for a specified setting distribution [31], is another objective; its maximization does not follow from a CGLMP value bound.

Appendix A: Proof of the negative-frequency bound

This appendix proves Lemma 2. The scalar function used in the proof is defined holomorphically on $|\mathrm{Re}\, z| < 1$ by

$$h(0) = \frac{8G}{\pi^2}, \qquad h'(0) = 0, \qquad h''(z) = -\sec\frac{\pi z}{2}.$$

On the real interval $-1 \le x \le 1$ it has the absolutely and uniformly convergent representation

$$h(x) = \frac{8}{\pi^2} \sum_{m \ge 0} \frac{(-1)^m}{(2m+1)^2} \cos\frac{(2m+1)\pi x}{2}. \tag{A1}$$

For real $|x| < 1$, the corresponding secant expansion is understood as the Abel limit

$$\sec(\pi x/2) = \lim_{r\uparrow 1} 2 \sum_{m\geq 0} (-1)^m r^{2m+1} \cos((2m+1)\pi x/2).$$

Integrating twice gives (A1), with $h(\pm 1) = 0$ and $h'(0) = 0$. Equation (A1) is not a complex-domain series: at any nonreal z its summands fail to tend to zero. The differential equation defines the holomorphic extension used below. For matrices whose spectra lie in the open strip, $h(X)$ is defined by holomorphic functional calculus, not substitution into a Taylor series about zero. In particular, $\tau_r h(X) = r^{-1} \sum_{\lambda\in\operatorname{spec} X} h(\lambda)$, with algebraic multiplicities; boundary values below are limits of the real part of this trace.

The mechanism is as follows. Lifting U to $\mathcal{W}(s) = sU(s^4)$ produces two commuting Hermitian involutions $A = \operatorname{Re}\mathcal{W} + \operatorname{Im}\mathcal{W}$ and $B = \operatorname{Re}\mathcal{W} - \operatorname{Im}\mathcal{W}$. Parseval's identity expresses $1 - 2\|P_- U\|_2^2$ as $\int \tau_r(A\mathsf{H}B)$, where H is the circular Hilbert transform. Pinching $\mathsf{H}B$ along B preserves this trace pairing. The positive pinching identity below compares the resulting commuting matrices with $B + \mathrm{i}\mathsf{H}B$, whose holomorphic trace can be averaged using its analytic extension to the disk. The boundary data $\operatorname{Re} h(\pm 1 + \mathrm{i}y) = |y|$ and the mean-value property then give the bound $h(0) = 8G/\pi^2$. The detailed proof includes both the strip-boundary and disk-boundary limiting arguments.

Details.—Throughout, B is a Hermitian matrix with $B^2 = I$ and Y is Hermitian. The *pinching* of Y along B,

$$\Pi_B(Y) = \frac{Y + BYB}{2},$$

keeps only the part of Y commuting with B; the matrix C above is $\Pi_B(\mathsf{H}B)$. The proof compares the matrix $B + \mathrm{i}Y$ to $B + \mathrm{i}\,\Pi_B(Y)$ through a nonnegative measure built from spectra of a non-Hermitian matrix pencil. The first lemma supplies the moments of that measure; it is Heinävaara's identity [29, Eq. (16), proof of Lemma 2.5], with $A - aI = H^{-1}$, his second matrix equal to Y, and the trace divided by r. The holomorphic pinching application is developed below. We write X_+ for the positive part of a Hermitian matrix X.

Lemma 7 (Trace-density moments)**.** *Let $H = H^\dagger$ be invertible and $Y = Y^\dagger$ in $M_r(\mathbb{C})$. Counting algebraic multiplicities, define*

$$w_{H,Y}(y) = \frac{1}{2\pi r} \sum_{\lambda\in\operatorname{spec}(H^{-1}(Y - yI))} |\operatorname{Im}\lambda|.$$

This function is continuous, nonnegative, and supported on the spectral interval of Y. For every integer $m \geq 0$,

$$\int y^m w_{H,Y}(y)\,\mathrm{d}y = \frac{1}{(m+1)(m+2)!} \times \frac{\mathrm{d}^{m+2}}{\mathrm{d}t^{m+2}} \tau_r (H + tY)_+^{m+1}\Big|_{t=0}. \tag{A2}$$

Proof. For $\operatorname{Im} z > 0$, an eigenvector v of $L(z) = H^{-1}(Y - zI)$ with eigenvalue λ satisfies $(Y - zI)v = \lambda Hv$; taking imaginary parts of $v^\dagger(Y - zI)v = \lambda\, v^\dagger Hv$ gives

$$-\operatorname{Im} z\, \|v\|^2 = \operatorname{Im}\lambda\; v^\dagger Hv.$$

Since H is invertible, the spectrum of $L(z)$ therefore avoids the real axis, and it splits into an upper and a lower cluster. The sum

$$g(z) = r^{-1} \sum_{\operatorname{Im}\lambda(L(z))>0} \lambda(L(z))$$

is holomorphic in the upper half-plane, being the trace of $L(z)$ times its upper-cluster Riesz projection, with algebraic multiplicities also at repeated or defective eigenvalues.

The sum extends continuously to the real axis. Choose disjoint small disks around the limiting eigenvalues. In a connected upper half-neighbourhood, the number of selected roots in each disk is constant. A real limiting cluster contributes an integer multiple of its real eigenvalue. At real y, the characteristic polynomial $\det(Y - yI - \lambda H)/\det H$ has real coefficients, so nonreal roots occur in conjugate pairs; half of each pair is selected, whence

$$\operatorname{Im} g(y + \mathrm{i}0) = \pi\, w_{H,Y}(y).$$

In particular $w_{H,Y}$ is continuous and nonnegative. Outside the spectral interval of Y, the matrix $Y - yI$ is definite, so the pencil is similar, up to sign, to the Hermitian matrix $(\pm(Y - yI))^{1/2} H^{-1} (\pm(Y - yI))^{1/2}$, and the density vanishes.

Near $w = 0$, put

$$K(w) = H^{-1}(I - wY), \qquad q(w) = \tau_r\big(K(w)\Pi_-(w)\big),$$

where $\Pi_-(w)$ continues the negative spectral cluster of H^{-1}. For small real w, similarity to $(I - wY)^{1/2} H^{-1} (I - wY)^{1/2}$ shows that q has real Taylor coefficients. In the upper exterior domain, $L(z) = -zK(1/z)$, and the selected upper cluster corresponds to the continued negative cluster, so $g(z) = -zq(1/z)$. Writing $q(w) = \sum_j q_j w^j$, Schwarz reflection and Cauchy's formula around the spectral slit give

$$g(z) = -q_0 z - q_1 + \int \frac{w_{H,Y}(y)}{y - z}\,\mathrm{d}y,$$
$$\int y^m w_{H,Y}(y)\,\mathrm{d}y = [w^{m+2}]\, q(w). \tag{A3}$$

The endpoint arcs vanish because pencil eigenvalues remain bounded near finite points; continuity controls the two banks of the slit.

It remains to identify the Taylor coefficients of q with the derivatives in (A2). Let $k = m + 2$ and $\mathsf{R}_H(\lambda) = (\lambda I - H)^{-1}$. Contours enclose the indicated positive or negative spectra and exclude zero. Expanding the logarithmic determinant ratio that generates the spectral cluster of $K(w)$ gives

$$\begin{aligned}
[w^k]\, q(w) &= \frac{(-1)^k}{kr\,2\pi\mathrm{i}} \oint_{\Gamma_{H^{-1},-}} \operatorname{Tr}[((\zeta H - I)^{-1}Y)^k]\,\mathrm{d}\zeta \\
&= -\frac{1}{kr\,2\pi\mathrm{i}} \oint_{\Gamma_{H,-}} \lambda^{k-2} \operatorname{Tr}[(\mathsf{R}_H(\lambda)Y)^k]\,\mathrm{d}\lambda \\
&= \frac{1}{kr\,2\pi\mathrm{i}} \oint_{\Gamma_{H,+}} \lambda^{k-2} \operatorname{Tr}[(\mathsf{R}_H(\lambda)Y)^k]\,\mathrm{d}\lambda.
\end{aligned}$$

The second line substitutes $\lambda = 1/\zeta$; the third uses the decay $O(\lambda^{-2})$ of the integrand at infinity to trade the negative contour for the positive one. Positive-cluster functional calculus and integration by parts, with

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\operatorname{Tr}[(\mathsf{R}_H(\lambda)Y)^k] = -k\operatorname{Tr}[(\mathsf{R}_H(\lambda)Y)^k\mathsf{R}_H(\lambda)],$$

give $[t^k]\,\tau_r(H+tY)_+^{k-1} = (k-1)\,[w^k]\,q(w)$. Together with (A3), this proves (A2). □

Lemma 8 (Positive pinching identity)**.** *Let $B = B^\dagger$ with $B^2 = I$, let $Y = Y^\dagger$, and put $C = \Pi_B(Y)$, $D = Y - C$. There is a nonnegative finite measure ν, supported on $[-1,1]$ times the spectral interval of Y, such that*

$$\begin{aligned}\tau_r f(sB+tY) - \tau_r f(sB+tC)\\ = t^2\int f''(-sx+ty)\,\mathrm{d}\nu(x,y)\end{aligned} \tag{A4}$$

for all polynomials f. Its density on $|x|<1$ is

$$\frac{\omega(x,y)}{1-x^2},\quad \omega(x,y) = \frac{1}{2\pi r}\sum_{\lambda\in\operatorname{spec}((B-xI)(Y-yI))}|\operatorname{Im}\lambda|, \tag{A5}$$

and

$$0 \le \frac{\omega(x,y)}{1-x^2} \le \frac{\|D\|}{2\pi\sqrt{1-x^2}},\qquad \nu(\mathbb{R}^2) = \frac{1}{2}\tau_r(D^2). \tag{A6}$$

Proof. Apply Lemma 7 with $H = (B-xI)^{-1}$, which is Hermitian and invertible for $|x|<1$, with the two eigenvalues $a = (1-x)^{-1} > 0$ and $b = -(1+x)^{-1} < 0$; note $H^{-1}(Y-yI) = (B-xI)(Y-yI)$, the pencil in (A5). Work in a basis diagonalizing B. For an index cycle $\mathbf{i} = (i_1,\dots,i_m)$, let p of its entries lie in the $+1$ eigenspace of B and $q = m-p$ in the -1 eigenspace, and put $Y_{\mathbf{i}} = Y_{i_1 i_2}\cdots Y_{i_m i_1}$. The contour form of (A2) evaluates by residues at $\lambda = a$:

$$\operatorname*{Res}_{\lambda=a}\frac{\lambda^{m-2}}{(\lambda-a)^p(\lambda-b)^q} = \binom{m-2}{p-1}\frac{a^{q-1}(-b)^{p-1}}{(a-b)^{m-1}} \tag{A7}$$

for $p,q\ge 1$, $p+q=m$. Substituting the values of a and b gives the moments of the slice density:

$$\begin{aligned}\int y^{m-2}\frac{\omega(x,y)}{1-x^2}\,\mathrm{d}y\\ = \frac{1}{rm\,2^{m-1}}\sum_{\mathbf{i}:\,1\le p<m}\binom{m-2}{p-1}\\ \times(1-x)^{p-1}(1+x)^{q-1}Y_{\mathbf{i}}.\end{aligned} \tag{A8}$$

Only mixed cycles, $1\le p<m$, contribute, because pure cycles have $p=0$ or $q=0$ and no residue at the opposite node.

Now expand the left side of (A4) for $f=\exp$. The coefficient of t^m in the mixed-cycle part of $\tau_r e^{sB+tY}$ is, by the Duhamel expansion,

$$\frac{1}{rm}\sum_{\mathbf{i}:\,1\le p<m}Y_{\mathbf{i}}\int_{\substack{u_j\ge 0\\ \sum_j u_j=1}} e^{s\sum_j b_{i_j}u_j}\,\mathrm{d}u,$$

where $b_i = \pm 1$ is the B-eigenvalue at index i. Pinching by B removes exactly the mixed cycles, so the left side of (A4) is the sum of these terms. Grouping the positive-B coordinates, whose total weight we call v, turns the simplex integral into a beta integral with density $v^{p-1}(1-v)^{q-1}/((p-1)!\,(q-1)!)$ against $e^{sv-s(1-v)}$. The substitution $x = 1-2v$, followed by (A8), gives the coefficient of t^m on the right of (A4), including the sign in $-sx$. Comparing coefficients of s and t proves the identity for $f=\exp$ and hence for all polynomials.

For the majorant in (A6), conjugate the pencil $(B-xI)(Y-yI)$ by the block-diagonal matrix

$$\operatorname{diag}(((1+x)/(1-x))^{1/4}I, ((1-x)/(1+x))^{1/4}I)$$

in the eigenbasis of B. The result has Hermitian diagonal blocks and off-diagonal blocks $\sqrt{1-x^2}\,D_{+-}$ and $-\sqrt{1-x^2}\,D_{+-}^\dagger$, so its anti-Hermitian part has norm at most $\sqrt{1-x^2}\,\|D\|$. By Bendixson's theorem the same bounds the imaginary part of each eigenvalue, proving the majorant. Integrating the $m=2$ case of (A8) over $x\in[-1,1]$ gives the total mass $\tau_r(D^2)/2$. □

Corollary 1 (Holomorphic pinching on a strip)**.** *Under the hypotheses of Lemma 8, identity* (A4) *holds with $s=\rho\in[0,1)$ and $t=\mathrm{i}$ for every f holomorphic on $|\operatorname{Re} z|<1$.*

Proof. Choose $\rho<\rho'<1$ and $L>\|Y\|$. The rectangle $\{z : |\operatorname{Re} z|\le\rho',\ |\operatorname{Im} z|\le L\}$ contains the spectra of $\rho B+\mathrm{i}Y$ and $\rho B+\mathrm{i}C$ and all points $-\rho x+\mathrm{i}y$ with (x,y) in the support of ν. Sum the polynomial identity for a large-parameter resolvent, continue in the connected exterior of this rectangle, and apply Cauchy's formula on its boundary. □

Proof of Lemma 2. Step 1: a function on the strip. On $|\operatorname{Re} z|<1$, define h by

$$h(0) = \frac{8G}{\pi^2},\qquad h'(0)=0,\qquad h''(z) = -\sec\frac{\pi z}{2};$$

explicitly $h'(z) = -\frac{2}{\pi}\log(\sec\frac{\pi z}{2}+\tan\frac{\pi z}{2})$, where the logarithm is the unique holomorphic branch on the strip whose value at $z=0$ is zero. Its real part $u(x,y) = \operatorname{Re} h(x+\mathrm{i}y)$ extends continuously to the closed strip and satisfies

$$u(\pm1,y) = |y|,\qquad 0\le u(x,y)\le\frac{8G}{\pi^2}+|y|. \tag{A9}$$

Indeed, on the boundary line $x=1$ one computes

$$-\operatorname{Im} h'(1+\mathrm{i}y) = \operatorname{sgn} y,$$

so $u(1,\cdot)$ has the slopes of $|y|$, and Taylor's formula with the classical integral $\int_0^1(1-x)\sec(\pi x/2)\,\mathrm{d}x = 8G/\pi^2$ gives $u(1,0)=0$. Since h is even, u is even in x, and $u_{xx} = -\operatorname{Re}\sec(\pi(x+\mathrm{i}y)/2)\le 0$: u is concave in x, hence at least its boundary values $|y|\ge 0$. On the centre line, $u(0,y) = 8G/\pi^2 + \int_0^y (y-s)\operatorname{sech}(\pi s/2)\,\mathrm{d}s \le 8G/\pi^2+|y|$ because $\int_0^\infty \operatorname{sech}(\pi s/2)\,\mathrm{d}s = 1$; concavity extends the upper bound to the strip.

For a matrix X with spectrum in the closed strip, put

$$\mathscr{U}(X) = r^{-1} \sum_{\lambda \in \operatorname{spec} X} \operatorname{Re} h(\lambda).$$

Apply Corollary 1 with $f = h$, take real parts, and let $\rho \uparrow 1$. Since $C = \Pi_B(Y)$ commutes with B, the matrix $B + \mathrm{i}C$ is normal with eigenvalues $\pm 1 + \mathrm{i}c$, where c runs over the spectrum of C on the two eigenspaces; by the boundary values in (A9), $\mathscr{U}(B + \mathrm{i}C) = \tau_r|\Pi_B(Y)|$. The limit therefore reads

$$\begin{aligned} &\mathscr{U}(B + \mathrm{i}Y) - \tau_r|\Pi_B(Y)| \\ &\qquad = \int \operatorname{Re} \sec \frac{\pi(x + \mathrm{i}y)}{2} \, \mathrm{d}\nu(x, y) \geq 0, \end{aligned} \tag{A10}$$

the integrand being nonnegative on the strip because $\cos(\pi z/2)$ maps it to the right half-plane. The passage $\rho \uparrow 1$ is dominated: each x-slice of the integral is bounded through (A6) by $\|D\|/(\pi\sqrt{1 - x^2})$, using $\int_{\mathbb{R}} \operatorname{Re} \sec(\pi(\rho x + \mathrm{i}y)/2) \, \mathrm{d}y = 2$; on the left side, restriction to smaller strips followed by passage to the endpoints proves convergence for the possibly nonnormal matrix $B + \mathrm{i}Y$.

Step 2: two commuting involutions. Set $\mathcal{W}(s) = sU(s^4)$ on the circle, so $\mathcal{W}$ is unitary with $\mathcal{W}^4 = I$, and

$$A = \operatorname{Re} \mathcal{W} + \operatorname{Im} \mathcal{W}, \qquad B = \operatorname{Re} \mathcal{W} - \operatorname{Im} \mathcal{W}.$$

Because the spectrum of $\mathcal{W}$ lies in $\{\pm 1, \pm \mathrm{i}\}$, the commuting Hermitian matrices $\operatorname{Re} \mathcal{W}$ and $\operatorname{Im} \mathcal{W}$ satisfy

$$(\operatorname{Re} \mathcal{W})(\operatorname{Im} \mathcal{W}) = 0 \quad \text{and} \quad (\operatorname{Re} \mathcal{W})^2 + (\operatorname{Im} \mathcal{W})^2 = I,$$

so A and B are commuting Hermitian involutions at every point of the circle. The identity $\mathcal{W}(\mathrm{i}s) = \mathrm{i}\,\mathcal{W}(s)$ shows that A and B have mean zero. If $U = \sum_k c_k z^k$, then $\mathcal{W} = \sum_k c_k s^{4k+1}$: the frequencies of $\mathcal{W}$ are $4k + 1$, with the same positive/negative split as those of U, and $\|P_- \mathcal{W}\|_2 = \|P_- U\|_2$.

Step 3: Parseval and the conclusion. Let H be the circular Hilbert transform, with multiplier $-\mathrm{i}\operatorname{sgn}(k)$ [32]. Expanding $\mathcal{W} = \frac{1}{2}[(1 + \mathrm{i})A + (1 - \mathrm{i})B]$ in Fourier coefficients and using the frequency split above, Parseval's identity gives

$$\begin{aligned} 1 - 2\|P_- U\|_2^2 &= \int_{\mathbb{T}} \tau_r(A \, \mathsf{H}B) \, \mathrm{d}m \\ &\leq \int_{\mathbb{T}} \tau_r|\Pi_B(\mathsf{H}B)| \, \mathrm{d}m. \end{aligned} \tag{A11}$$

The inequality holds pointwise: pinching is self-adjoint for the trace pairing and fixes A, since $BAB = AB^2 = A$; hence $\tau(A \, \mathsf{H}B) = \tau\big(A \, \Pi_B(\mathsf{H}B)\big) \leq \tau|\Pi_B(\mathsf{H}B)|$, because $\|A\| \leq 1$.

The analytic matrix function with boundary values $B + \mathrm{i}\mathsf{H}B$ vanishes at the disk centre, since B has mean zero. The real part of the holomorphic trace of h of this analytic matrix family is harmonic, so the mean-value property gives

$$\int_{\mathbb{T}} \mathscr{U}(B + \mathrm{i}\mathsf{H}B) \, \mathrm{d}m = \mathscr{U}(0) = \frac{8G}{\pi^2}.$$

For the boundary passage, use radial limits together with the bound $0 \leq \mathscr{U}(X) \leq 8G/\pi^2 + \tau_r|\operatorname{Im} X|$, which follows from (A9), Schur triangularization, and convexity of the absolute value on the diagonal; the radial Hilbert transforms are uniformly L^2-bounded, hence the displayed quantities are uniformly integrable. Applying (A10) pointwise in (A11) now gives $1 - 2\|P_- U\|_2^2 \leq 8G/\pi^2$, which is (19). □

Appendix B: The circulation sum of squares

This appendix proves (31). Fix $i < j < k$, and write

$$\begin{aligned} u_\ell &= E_{i\ell} E_{j,\ell+2}, \;\; v_\ell = E_{j\ell} E_{k,\ell+2}, \;\; z_\ell = E_{i\ell} E_{k,\ell+2}, \\ a_\ell &= E_{i\ell} E_{j,\ell+1}, \;\; b_\ell = E_{j\ell} E_{k,\ell+1}, \;\; c_\ell = E_{i\ell} E_{k,\ell+1}, \end{aligned}$$

the forbidden products on the three edges of the triangle. Let P be the sum of the squared norms $\|\cdot\|_2^2$ of all a_ℓ, b_ℓ, c_ℓ, and O the corresponding sum for u_ℓ, v_ℓ, z_ℓ, so that $P + O = p_{ij} + p_{jk} + p_{ik}$ by (10).

TABLE I. Terms in the circulation identity. Spectral indices are read modulo four.

$X_{\ell\nu}$	$Y_{\ell\nu}$	ϵ_ν
z_ℓ	a_ℓ	1
$z_\ell^\dagger$	$b_{\ell+1}^\dagger$	1
$v_{\ell+1}$	$a_\ell^\dagger$	1
u_ℓ	c_ℓ	−1
$u_\ell^\dagger$	$b_{\ell+2}$	1
$v_{\ell+3}^\dagger$	$c_\ell^\dagger$	−1

Projection completeness ($\sum_m E_{jm} = I$, inserted between the two factors of z_ℓ and c_ℓ) gives

$$\begin{aligned} C_{ijk} = -2 \sum_\ell \|z_\ell\|_2^2 + 2 \operatorname{Re} \sum_\ell \big[\langle u_\ell, z_\ell \rangle + \langle v_\ell^\dagger, z_\ell^\dagger \rangle \big] \\ + 2 \operatorname{Re} \sum_{\ell, \nu} \epsilon_\nu \langle X_{\ell\nu}, Y_{\ell\nu} \rangle, \end{aligned} \tag{B1}$$

with the six triples $(X_{\ell\nu}, Y_{\ell\nu}, \epsilon_\nu)$ of Table I. For $\sigma = \pm 1$, completing squares yields the identity

$$\begin{aligned} 2P + 5O - \sigma C_{ijk} = &\sum_{\ell, \nu} \|Y_{\ell\nu} - \sigma\epsilon_\nu X_{\ell\nu}\|_2^2 \\ &+ 3 \sum_\ell \|u_\ell - \tfrac{\sigma}{3} z_\ell\|_2^2 \\ &+ 3 \sum_\ell \|v_\ell^\dagger - \tfrac{\sigma}{3} z_\ell^\dagger\|_2^2 \\ &+ \Big(\frac{7}{3} + 2\sigma \Big) \sum_\ell \|z_\ell\|_2^2 \;\geq\; 0, \end{aligned} \tag{B2}$$

whence $|C_{ijk}| \leq 2P + 5O \leq 5(P + O)$, which is (31). Both (B1) and (B2) are finite identities in the twelve projections: they are verified by expanding $E_{j3} = I - E_{j0} - E_{j1} - E_{j2}$ at each site, reducing adjacent same-site products by orthogonality, and comparing coefficients of cyclic words—choosing the lexicographically least cyclic rotation of each word and splitting each real part into the half-sum of a word and its adjoint.

Appendix C: Bounds on the error coefficients

This appendix proves Lemma 6. Throughout, $y = 1 - k/d$, and

$$r_{\mathrm{reg}}(y) = \csc\frac{\pi y}{2} - \frac{2}{\pi y}, \qquad \varphi(y) = y(1-y)\, r_{\mathrm{reg}}(y).$$

The cosecant expansion gives

$$\begin{aligned} r_{\mathrm{reg}}(y) &= \sum_{n\ge 1} r_n y^{2n-1}, \\ r_n &= \frac{2(2^{2n-1}-1)|B_{2n}|}{(2n)!}\Big(\frac{\pi}{2}\Big)^{2n-1}, \end{aligned} \tag{C1}$$

with Bernoulli numbers B_{2n} and $0 < r_n < 3\cdot 4^{-n}$. Since $\sigma(x) = \csc(\pi y/2)$ and $1 - y = x$, one checks $x(1-x)\sigma(x) - 2x/\pi = \varphi(y)$, so the kernel b of (25) is $b = -\varphi''(y)/2$. Here $x(1-x)$ already cancels the secant pole; subtracting $2x/\pi$ leaves the contribution of the regular part r_{reg}, with zero endpoint values. The Bernstein averages themselves are beta expectations: with $X \sim \mathrm{Beta}(k+1, d-k+1)$ and $Y = 1 - X$, $(B_d g)(k) = \frac{d}{d+1}\,\mathbb{E}[g(X)]$, the moments are

$$\frac{d}{d+1}\,\mathbb{E}[Y^j] = m_j := \frac{d}{d+1}\,\frac{(d-k+1)_j}{(d+2)_j},$$

and $\mathbb{E}[Y^{-1}] = (d+1)/(d-k)$, which cancels the pole of $\sigma(k/d)$ in (25). Substituting (C1) term by term gives the absolutely convergent series

$$\begin{aligned} e_{d,k} = \sum_{n\ge 1} r_n \Big[y^j - \Big(1 + \frac{(j+1)(j+2)}{2d}\Big) m_j \\ + \frac{j(j+1)}{2d} m_{j-1} \Big], \end{aligned} \tag{C2}$$

with $j = 2n-1$. Each bracket is bounded in absolute value by $2 + 4n^2/(d+1)$, since $y^j, m_j \in [0,1]$ and $m_j \le d/(d+1)$.

1. The beta–Stein estimate: proof of (i)

Let $Z \sim \mathrm{Beta}(dy, d(1-y)) = \mathrm{Beta}(d-k, k)$, so $\mathbb{E}\,Z = y$. Multiplying the $\mathrm{Beta}(d-k,k)$ density by $z(1-z)$ reproduces the $\mathrm{Beta}(d-k+1, k+1)$ kernel, which turns the Bernstein average into a size-biased expectation, $(B_d g)(k) = \mathbb{E}[Z(1-Z)g(1-Z)]/[y(1-y)]$, and integration by parts against the beta density gives the Stein identity [33, 34]

$$\mathbb{E}[(Z-y)g(Z)] = \frac{1}{d}\,\mathbb{E}[Z(1-Z)g'(Z)].$$

Combining the two with (25) yields the exact representation

$$e_{d,k} = \frac{\varphi(y) - \mathbb{E}\,\varphi(Z) + \frac{1}{2}\,\mathbb{E}[(Z-y)\varphi'(Z)]}{y(1-y)}. \tag{C3}$$

The numerator is a trapezoidal remainder: since

$$\mathbb{E}[(Z-y)\varphi'(y)] = 0$$

may be added freely, with $\xi = Z - y$,

$$\varphi(y) - \varphi(y+\xi) + \frac{\xi}{2}[\varphi'(y+\xi) + \varphi'(y)] = \frac{\xi^3}{12}\varphi'''(y) + \epsilon_\xi,$$

where $|\epsilon_\xi| \le \|\varphi^{(4)}\|_\infty |\xi|^4/24$. The derivatives of φ are controlled by the positive series (C1): for $j = 2n-1$, the third and fourth derivatives of $y^{j+1} - y^{j+2}$ are bounded on $[0,1]$ by $3j(j+1)$ and $4j(j-1)(j+1)$, and the sums

$$r''_{\mathrm{reg}}(1) + 2r'_{\mathrm{reg}}(1) = \frac{\pi^2}{4}, \qquad r'''_{\mathrm{reg}}(1) + 3r''_{\mathrm{reg}}(1) = \frac{3\pi^2}{4}$$

give $\|\varphi'''\|_\infty \le 3\pi^2/4$ and $\|\varphi^{(4)}\|_\infty \le 3\pi^2$. The central beta moments are

$$\mathbb{E}\,\xi^3 = \frac{2y(1-y)(1-2y)}{(d+1)(d+2)},$$

$$\mathbb{E}\,\xi^4 = \frac{3y(1-y)[2(1-2y)^2 + y(1-y)(d+2)]}{(d+1)(d+2)(d+3)}.$$

Inserting these into (C3), with $q = |1-2y|$,

$$|e_{d,k}| \le \frac{\pi^2}{8(d+1)(d+2)}\Big[q + \frac{3(d+2)}{4(d+3)} - \frac{3(d-6)}{4(d+3)}q^2\Big].$$

For $d \ge 66$, maximizing the bracket over $q \in [0,1]$ gives at most

$$\frac{3(d+2)}{4(d+3)} + \frac{d+3}{3(d-6)} < \frac{9}{8},$$

the difference from $9/8$ being

$$(d^2 - 57d - 342)/[24(d-6)(d+3)] > 0.$$

Using $\pi^2 < 10$ gives (36).

2. The interval computation: proof of (ii)

For $2 \le d \le 65$, the inequality (37) is a finite set of rational comparisons, verified in exact interval arithmetic [35] as follows. Truncating (C2) after 32 terms has error at most

$$T_d = 3\sum_{n=33}^{\infty} 4^{-n}\Big(2 + \frac{4n^2}{d+1}\Big), \tag{C4}$$

evaluated exactly with

$$\begin{aligned} \sum_{n\ge J} q^n &= \frac{q^J}{1-q}, \\ \sum_{n\ge J} n^2 q^n &= q^J\Big[\frac{J^2}{1-q} + \frac{2Jq}{(1-q)^2} + \frac{q(1+q)}{(1-q)^3}\Big] \end{aligned}$$

at $q = 1/4$. For the coefficients, put

$$A(a,N) = \sum_{i=0}^{N-1} (-1)^i/[(2i+1)a^{2i+1}];$$

Machin's identity gives the rational enclosure

$$\pi \in 16[A(5,128), A(5,129)] - 4[A(239,128), A(239,129)].$$

Generate the Bernoulli numbers by $B_0 = 1$ and $B_m = -(m+1)^{-1}\sum_{i=0}^{m-1}\binom{m+1}{i}B_i$ through $m = 64$, and substitute the enclosure of π into (C1). The moments follow the recursion $m_0 = d/(d+1)$, $m_j = m_{j-1}(d-k+j)/(d+j+1)$. Endpoint inclusion in these rational operations, followed by adding $[-T_d, T_d]$, encloses each $e_{d,k}$; the skew coefficients $t_k = e_{d,d-k} - e_{d,k}$ and the loads $\ell_{d,k}$ of (34) are enclosed likewise, bounding each absolute value $|[l,u]|$ by $\max(|l|,|u|)$. The parameters are collected in Table II: over all $2 \le d \le 65$ and $1 \le k < d$, the normalized margin satisfies

$$(d+1)\Big[\frac{1}{3(d+1)} + \min_k e_{d,k} - \frac{5}{2d}\max_k \ell_{d,k}\Big] \ge \frac{1935}{10^4} > \frac{1}{6},$$

which is (37). The accompanying `verification/verify_coefficients.py` checks all 2,080 pairs using Python 3 and only its standard library. Run `python3 verification/verify_coefficients.py` from the manuscript directory. The worst dimension is $d = 7$, and the certified minimum normalized margin exceeds 0.19354436777798498. The Machin enclosure is rounded outward to 80 decimal places and the r_n enclosures to 60; all rounding, propagation, and comparisons use exact rational numbers. The companion script `verification/verify_circulation.py` checks (B1) and both signs of (B2) by exact cyclic-word arithmetic. Commands, provenance, expected output, and recorded outputs accompany the scripts in `verification/`. These computations certify the stated finite inequalities and identities, not the full analytic proof.

TABLE II. Data for the finite interval computation. The margin uses lower endpoints for $e_{d,k}$ and upper bounds for $\ell_{d,k}$.

Quantity	Data
Outcome counts	$2 \le d \le 65$
Separations	$1 \le k < d$
Cosecant terms	$1 \le n \le 32$
Moment indices	$0 \le j \le 63$
Bernoulli indices	$0 \le m \le 64$
Arctangent partial sums	128, 129 terms
Outward rounding (π, r_n)	80, 60 decimal places
Worst dimension	$d = 7$
Tail start in (C4)	$J = 33$
Lower bound for the normalized margin	$1935/10000 > 1/6$

Appendix D: Arbitrary local dimension

Problem 27.B concerns the maximally entangled state of two d-level systems, which is Theorem 1. The proof, however, works on Φ_n for every n until its very last step, and yields the following extension.

Theorem 3 (General local dimension)**.** *Let $d \ge 2$ and $n \ge 1$. Every projective strategy on Φ_n satisfies $I_d \le I_F(d)$. Equality holds if and only if d divides n and there is a unitary U on $\mathbb{C}^n = \mathbb{C}^d \otimes \mathbb{C}^{n/d}$ such that the tracial PVMs are $P_a^t = U(\Pi_a^t \otimes I_{n/d})U^\dagger$ for all t, a. In particular, if d does not divide n, the value $I_F(d)$ is not attained on Φ_n.*

Proof. The inequality was proved for every n at the end of Section III. Suppose equality. Steps 1–3 of the proof of Theorem 1 make no use of $n = d$ and give a unitary W on $\mathbb{C}^d \otimes \mathbb{C}^{4n}$ with $\mathcal{M}_a^t(V) = W(\Pi_a^t \otimes I_{4n})W^\dagger$ for all t, a. As in Step 4 there, the commutant argument gives $W^\dagger \Lambda W = I_d \otimes p$ for the range projection $\Lambda = \mathcal{Z}\mathcal{Z}^\dagger$ of the isometry (15) and a projection p on $\mathbb{C}^{4n}$. Comparing ranks, $n = \operatorname{rank}\Lambda = d \operatorname{rank} p$, which forces $d \mid n$. Writing $k = n/d$ and choosing an isometry $\iota : \mathbb{C}^k \to \mathbb{C}^{4n}$ onto the range of p, the unitary $T = (I_d \otimes \iota^\dagger)W^\dagger \mathcal{Z} : \mathbb{C}^n \to \mathbb{C}^d \otimes \mathbb{C}^k$ satisfies

$$TP_a^t T^\dagger = (I_d \otimes \iota^\dagger)(\Pi_a^t \otimes p)(I_d \otimes \iota) = \Pi_a^t \otimes I_k,$$

the claimed classification with $U = T^\dagger$.

Conversely, an amplified strategy $P_a^t = U(\Pi_a^t \otimes I_k)U^\dagger$ on Φ_{dk} has the same behaviour as the standard strategy on Φ_d, since $\tau_{dk}\big((\Pi \otimes I_k)(\Pi' \otimes I_k)\big) = \tau_d(\Pi\Pi')$, and therefore attains $I_F(d)$. □

For fixed d, n, the space of four d-outcome PVMs on $\mathbb{C}^n$ is compact: it is a closed, bounded subset of a finite-dimensional matrix space. The Bell functional is continuous, so its optimum is attained. The equality classification therefore gives a strictly positive gap when $d \nmid n$, without specifying a rate. Write $s = n \bmod d > 0$. Pad $\lfloor n/d \rfloor$ standard blocks with s one-dimensional blocks on which all four PVMs assign one fixed outcome achieving the local value 2. The resulting behaviour is the mixture of the standard behaviour, with weight $d\lfloor n/d \rfloor/n$, and a deterministic local behaviour, with weight s/n, so that

$$\sup\{I_d \text{ on } \Phi_n\} \;\ge\; I_F(d) - \frac{s}{n}\big(I_F(d) - 2\big).$$

The deficit of the optimum is therefore at most of order $1/n$ along every residue class; lower bounds on it require a quantitative version of the rigidity argument and are not pursued here.

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